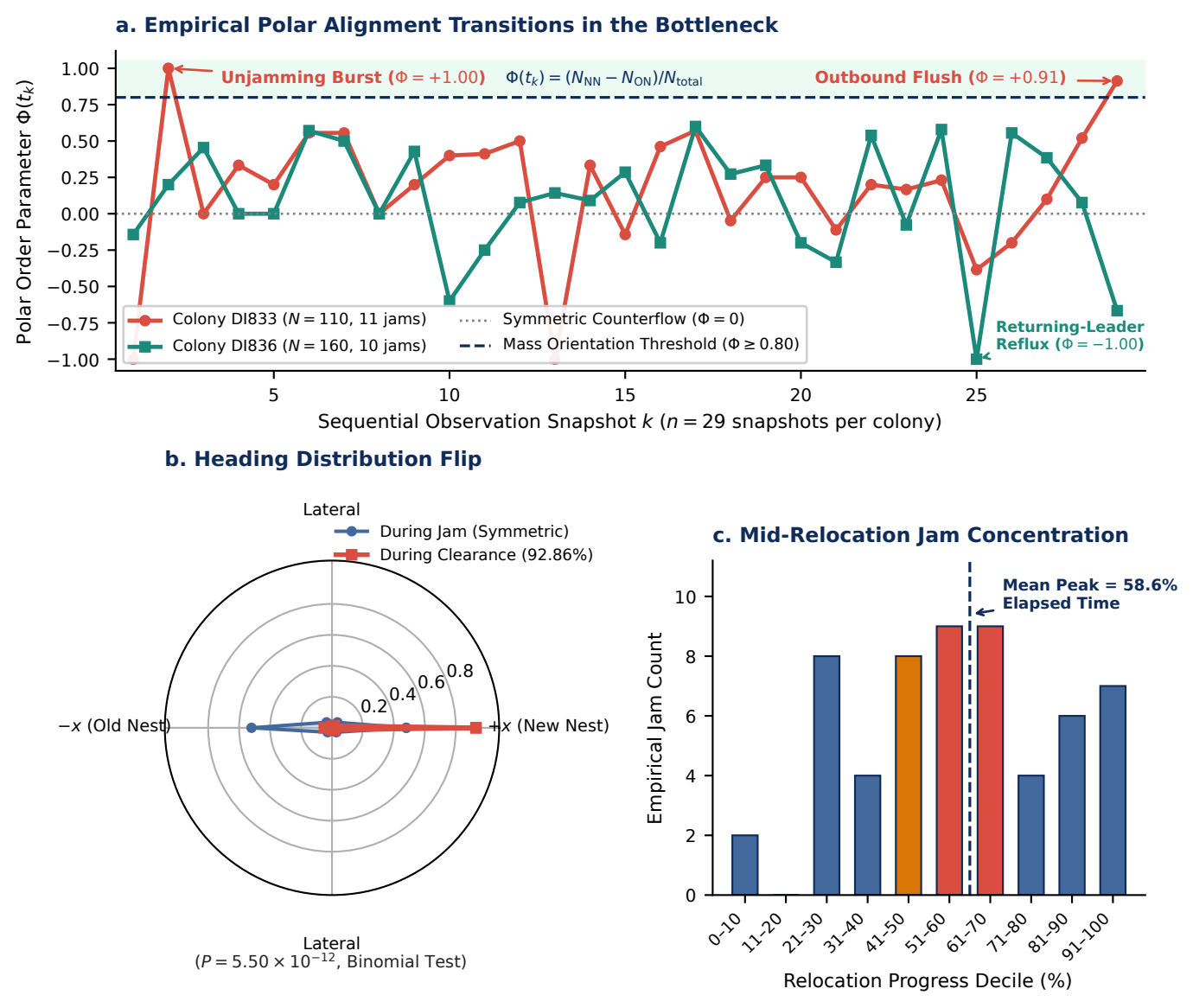
a. Empirical Polar Alignment Transitions in the Bottleneck
Unjamming Burst (Φ = +1.00)
Φ(t_k) = (N_NN − N_ON)/N_total
Outbound Flush (Φ = +0.91)
Polar Order Parameter Φ(t_k)
1.00
0.75
0.50
0.25
0.00
−0.25
−0.50
−0.75
−1.00
Colony DI833 (N = 110, 11 jams)
Colony DI836 (N = 160, 10 jams)
Symmetric Counterflow (Φ = 0)
Mass Orientation Threshold (Φ ≥ 0.80)
Returning-Leader Reflux (Φ = −1.00)
5
10
15
20
25
Sequential Observation Snapshot k (n = 29 snapshots per colony)
b. Heading Distribution Flip
Lateral
During Jam (Symmetric)
During Clearance (92.86%)
0.2
0.4
0.6
0.8
−x (Old Nest)
+x (New Nest)
Lateral
(P = 5.50 × 10^−12, Binomial Test)
c. Mid-Relocation Jam Concentration
Mean Peak = 58.6% Elapsed Time
Empirical Jam Count
0
2
4
6
8
10
0–10
11–20
21–30
31–40
41–50
51–60
61–70
71–80
81–90
91–100
Relocation Progress Decile (%)